# Statistical Process Control — I-MR and X-bar/R

*Battery Cycling Quality Analytics — Analyze Phase*

## Method

Baseline control limits computed from cycles 1 to 30 for each cell. I-MR chart: individuals and moving range. Sigma estimated from average moving range divided by d2 = 1.128. X-bar / R chart: subgroup size = 5 cycles. All 4 Western Electric rules applied to each I-chart.

## Per-cell baseline + violation summary

| Cell | Baseline X■ (Ah) | σ■ (Ah) | UCL (Ah) | LCL (Ah) | Total points | Violations |
|---|---|---|---|---|---|---|
| B0005 | 1.8357 | 0.00307 | 1.8449 | 1.8265 | 168 | 595 |
| B0006 | 1.9884 | 0.01029 | 2.0193 | 1.9575 | 168 | 629 |
| B0007 | 1.9000 | 0.00250 | 1.9075 | 1.8924 | 168 | 602 |
| B0018 | 1.8420 | 0.00721 | 1.8637 | 1.8204 | 132 | 489 |

## Interpretation

Every cell shows a large number of post-baseline violations. This is expected and informative: SPC limits built from the first 30 cycles represent the stable new-cell operating window. As capacity fade progresses, the signal intentionally breaks the LCL — the SPC chart is detecting the aging process itself. The baseline window is the 'in-control' reference; violations after cycle 30 map directly onto the fade trajectories quantified in Notebook 05. For a production Quality Engineer, this chart would trigger investigation the moment a specific cell crosses the LCL earlier than the fleet average.

## Verdict

SPC baseline established. Fade behavior cleanly separates from measurement noise. Chart is ready for use in Cp / Cpk capability study (Notebook 10) and as reference for Design of Experiments in Notebook 09.

